Step 2a - H8 - Higher ore extraction yields

Market scan

Market size

The value pool is mineral recovery inside the ~$23B/yr global mineral processing equipment market, but the real prize is recovered metal: a 1% recovery uplift on a large copper concentrator is worth roughly $5-15M/yr in extra revenue — far more than the equipment cost. Liberation/recovery improvements target flotation plants and leach operations within the same comminution-adjacent spend (~$10-15B/yr on crushing/grinding equipment plus separation). Demand is amplified by declining ore grades industry-wide, which forces finer grinding and makes liberation the bottleneck of recovery.

Region	Size ($B)	Region	Size ($B)
Europe	~$2-3B/yr (~10%; Scandinavia + new CRMA-driven projects)	Asia	~$9-10B/yr (~40-45%; China, Australia, India concentrator fleets)
America	~$5-6B/yr (~25%; Chile/Peru copper + North American gold/copper)	Africa	~$3-4B/yr incl. Middle East (~15%; copper/cobalt/gold growth, ~10% CAGR)

Target customers

Same customer base as ore weakening — hard-rock concentrators and leach operations, ~3,000-5,000 sites of the ~15,000 facilities in ICMM's dataset — but the value case differs: recovery uplift pays best on high-value ores (gold, copper, PGM, refractory and low-grade deposits where every recovery point matters). Large concentrators monetise +1% recovery immediately; refractory-gold and heap-leach operators are a particularly receptive niche; small mines need simple bolt-on units. Sales motion is via metallurgical testwork → plant trial, typically 1-3 year adoption cycles.

Customer size	Approx. number	Market share (in %)
Large	~200-400 concentrators (>10 Mt/y; majors, high revenue per recovery point)	~70-80% of world processed tonnage
Medium	~1,000-1,500 concentrators incl. gold/refractory niche (1-10 Mt/y)	~15-25% of tonnage
Small	~3,000+ sites (<1 Mt/y; heap-leach and gravity gold)	<10% of tonnage

Players offering a solution today

Recovery performance is today delivered by the same concentrated OEM tier (Metso, FLSmidth, Weir — #1 positions in flotation/classification) plus reagent/leaching specialists; incumbents optimise recovery inside conventional flowsheets rather than changing ore breakage physics. No commercial liberation-enhancing pre-treatment supplier exists: the emerging field (Nottingham/e2v MicroHammer microwave pilots; SELFRAG high-voltage pulse machines with JKMRC/Newcrest testwork showing selective grain-boundary breakage) is pre-commercial, leaving the liberation pre-treatment space open.

Player	Type of solution	Market share (in %)

Large (Metso, FLSmidth, Weir)	Flotation cells, classifiers/cyclones, regrind mills, hydrometallurgy — recovery via conventional flowsheet optimisation; strong aftermarket lock-in	top 3 ~35-50% of relevant equipment; Metso #1 flotation/filtration
Medium (Eriez, TAKRAF/Delkor, Tenova, Multotec, Outotec-alumni specialists)	Separation, leaching, thickening, specialist recovery equipment	~20-30%
Small/emerging (SELFRAG HVP, Teledyne e2v MicroHammer, Lightning Machines, TOMRA/Steinert sorting)	Liberation-enhancing pre-treatment: pulse fragmentation, microwave pilots, sensor-based pre-concentration	<5%, fragmented, pre-commercial

Regulation

Regulation	Jurisdiction	What it means for us
EU Critical Raw Materials Act (2024)	EU	Targets 10% domestic extraction / 40% processing by 2030 and defines strategic recycling/recovery benchmarks — recovery-maximising technology aligns with EU resource-efficiency objectives and can ease project permitting narratives.
EU Industrial Emissions Directive / MIN BREF	EU	BAT conclusions cover beneficiation/recovery efficiency and specific consumption indicators; better liberation at coarser grind reduces energy AND reagent/water intensity, supporting compliance.
EU ETS + Energy Efficiency Directive Art. 11 audits	EU	Coarser-grind-for-equal-liberation cuts grinding energy; audited energy savings plus avoided metal losses both improve the economics regulators increasingly scrutinise.
Tailings regulations (EU Extractive Waste Directive; GISTM global standard)	EU/global	Higher recovery = less metal and sulphide to tailings; post-Brumadinho GISTM compliance pressure favours technologies reducing tailings volumes and sulphide content.
Australia Safeguard Mechanism / ICMM net-zero commitments	Australia/global	Declining grades raise energy per tonne of metal; recovery uplift lowers CO2 per unit metal — supports decarbonisation plans of majors committed to net zero.

EMF/RF safety: CISPR 11/EN 55011 ISM + EU Directive 2013/35/EU	EU/international	Microwave units require shielding to ISM leakage limits (915 MHz/2.45 GHz) and worker EMF limits — engineering requirement, not a licensing barrier.

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
Name	Devin (agent draft)	Devin (agent draft)	Date	2026-10-30	2026-10-30	Name			Date	
Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis